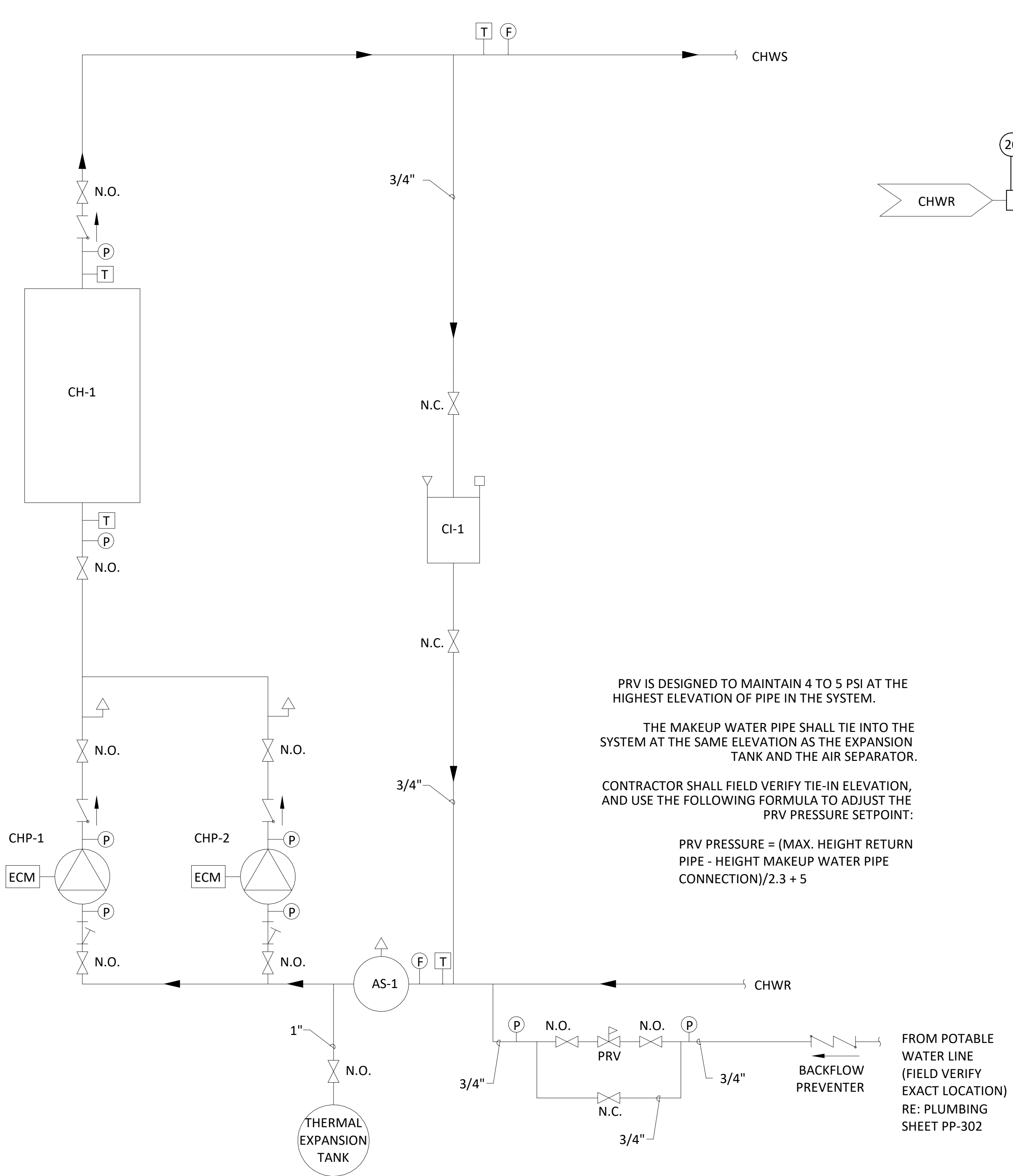


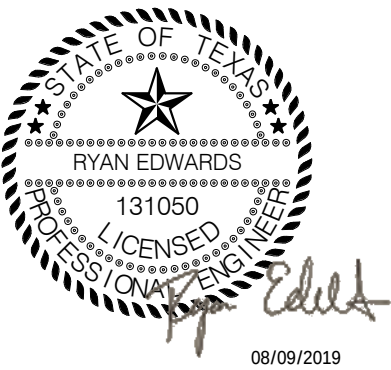
A
B
C
D



1 CHILLED WATER SYSTEM P&ID
NOT TO SCALE

BAS INPUT/OUTPUT POINT LIST						
TAG	POINT NAME	AI	AO	DI	DO	ALARM
1	CHILLER 1 STATUS*			X		
2	CHILLER 1 ENABLE/DISABLE				X	
3	CHILLER 1 FAILURE*					X
4	CHILLER 1 LOAD*	X				
5	HIGH/LOW CHW PRESSURE DROP ACROSS CHILLER 1	X				X
6	CHILLER 1 PROOF OF FLOW*			X		X
7	CHILLER 1 LOW CHW FLOW ALARM*					X
8	CHP 1 STATUS VIA CT			X		
9	CHP 1 ON/OFF				X	
10	CHP 1 FAILURE					X
11	CHP 1 SPEED		X			
12	CHP 1 ECM FAULT					X
13	CHP 2 STATUS VIA CT			X		
14	CHP 2 ON/OFF				X	
15	CHP 2 FAILURE					X
16	CHP 2 SPEED		X			
17	CHP 2 ECM FAULT					X
18	CHILLED WATER SUPPLY TEMPERATURE*	X				
19	CHILLER WATER SUPPLY TEMPERATURE SETPOINT		X			
20	CHILLED WATER RETURN TEMPERATURE*	X				
21	CHILLED WATER SYSTEM BYPASS VALVE POSITION		X			
22	OUTSIDE AIR TEMPERATURE (GLOBAL)	X				
23	EMERGENCY SHUTDOWN (GLOBAL)			X		X
24	CHILLED WATER FLOW RATE (GPM)	X				
25	FREEZE PROTECTION					X

* INDICATES POINT PULLED FROM CHILLER CONTROLLER



US Army Corps
of Engineers ®

DESIGNED BY: Designer		ISSUE DATE: AUGUST 12, 2019	
DRAWN BY: Draftsman		SOLICITATION NO.: 01120315B206	
CHECKED BY: Checker		CONTRACT NO.:	
SUBMITTED BY:		FILE NUMBER:	
FILE NAME: ANSI'D' KF317593.BLDG615 HVAC		SIZE:	

US ARMY CORPS OF ENGINEERS
JOINT BASE SAN ANTONIO
FORT SAM HOUSTON, TX

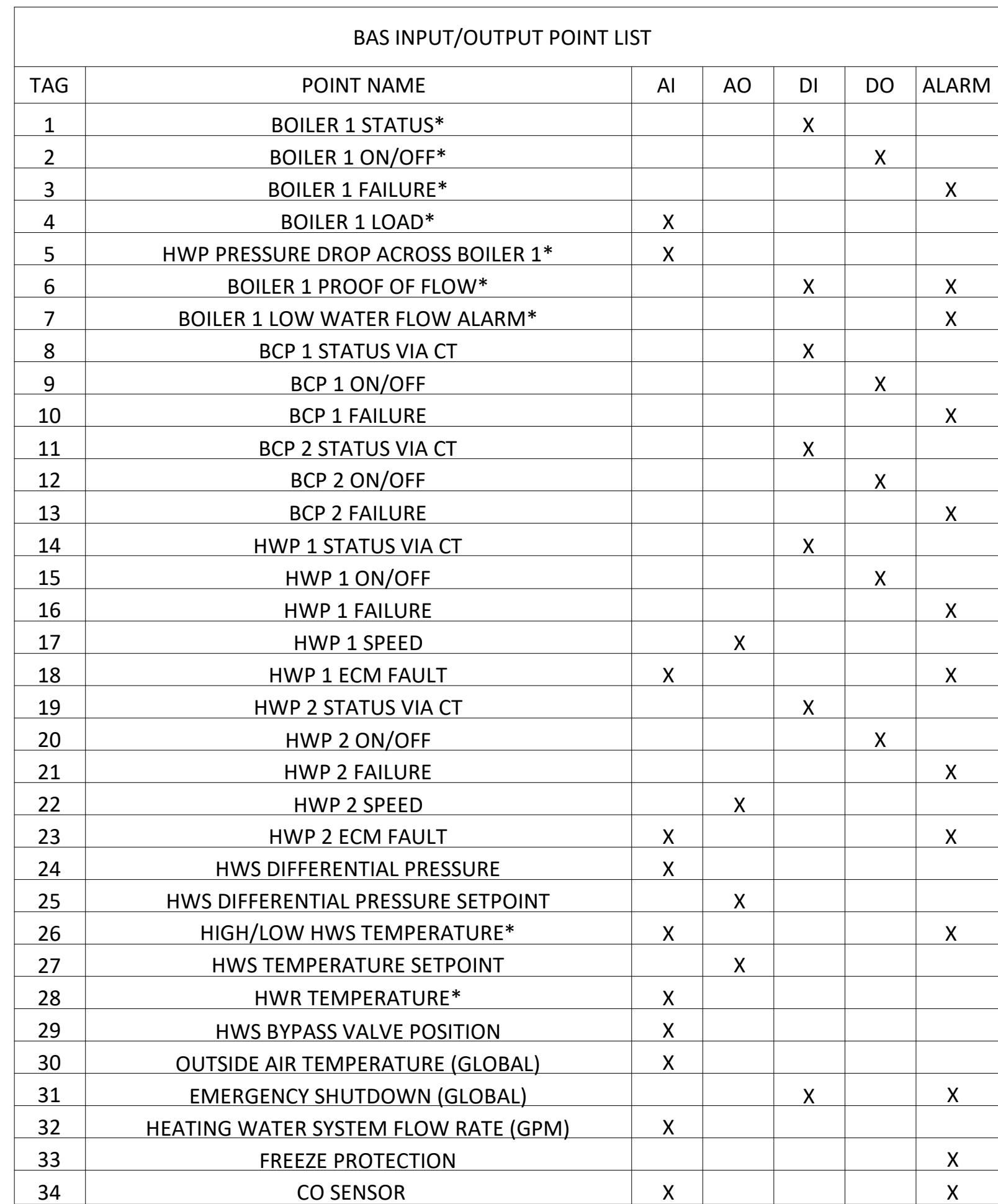
KENALL + FNI
6200 SAVOY DRIVE, SUITE 600
HOUSTON, TEXAS 77066
PHONE: (832) 251-4200

RENOVATION OF BUILDING 615

MECHANICAL CONTROL SCHEMATICS II

SHEET ID

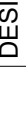
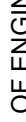
M-507



NOT TO SCALE

NOT TO SCALE

[illegible]

US ARMY CORPS OF ENGINEERS JOINT BASE SAN ANTONIO FORT SAM HOUSTON, TX	 +  KERNELL & FNI	DESIGNED BY:	ISSUE DATE:
		Designer DRAWN BY: AutoCAD CHECKED BY: C:\	AUGUST 12, 2019 SOLICITATION NO.: W9126G1952386 CONTRACT NO.:
6200 SAVOY DRIVE, SUITE 600 HOUSTON, TEXAS 77036 PHONE: (832) 251-1200		SUBMITTED BY:	FILE NUMBER:
		SIZE: ANSI/D: K317593, B1 D6615 HVAC	FILE NAME:

RENOVATION OF BUILDING 615

MECHANICAL CONTROL SCHEMATICS IV

SHEET ID

M-509



C

FAN COIL UNIT (FCU) CONTROL SEQUENCE

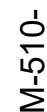
- B**

A

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FCU P&ID

NOT TO SCALE

[illegible]

US ARMY CORPS OF ENGINEERS JOINT BASE SAN ANTONIO FORT SAM HOUSTON, TX	 + 	DESIGNED BY: Designer	ISSUE DATE: AUGUST 12 1919
		DRAWN BY: N912631953286	SOLICITATION NO.: N912631953286
		CHECKED BY: Checker	CONTRACT NO.: N912631953286
KENALL + FNI 6200 SAVOY DRIVE, SUITE 600 HOUSTON, TEXAS 77068 PHONE: (832) 251-5200		SUBMITTED BY: Designer	FILE NUMBER: N912631953286
		SIZE: ANSI D	FILE NAME: KF37593 BLDG616 H/VAC

RENOVATION OF BUILDING 615

MECHANICAL CONTROL SCHEMATICS V

SHEET ID

M-510

FINAL SUBMITTAL





1. DOAS:
 - 1.1 THE DOAS SHALL BE A PACKAGED UNIT WITH A SUPPLY FAN AND EXHAUST FAN.
 - 1.2 THE DOAS SHALL BE PROVIDED WITH A VFD FOR THE SUPPLY FAN, AND A VFD FOR THE EXHAUST FAN.
 - 1.3 THE DOAS SHALL BE CONTROLLED BY THE BAS AND REMAIN ENABLED AT ALL TIMES.
 - 1.3.1 DURING OCCUPIED HOURS, THE DOAS SHALL RUN CONTINUOUSLY AND DELIVER A CONSTANT AIR VOLUME CALCULATED PER ASHRAE 62.1.
 - 1.3.2 DURING UNOCCUPIED HOURS, THE DOAS SHALL NOT RUN UNLESS A ZONE OVERRIDE COMMAND IS RECEIVED.
 - 1.3.3 OCCUPIED/UNOCCUPIED HOURS SHALL BE DETERMINED BY BUILDING SCHEDULE IN THE BAS.
2. DOAS CONTROL: THE DOAS SYSTEM SHALL BE PROVIDED WITH A FACTORY FURNISHED ON-BOARD BACNET CONTROLLER THE MANUFACTURER SHALL USE ITS PREFERRED SEQUENCE TO MEET THE FOLLOWING REQUIREMENTS:
 - 2.1 IN COOLING, THE CONTROLLER SHALL MONITOR THE CHILLED WATER COIL LEAVING AIR TEMPERATURE AND MODULATE CHILLED WATER FLOW TO MAINTAIN THE CHILLED WATER COIL LEAVING AIR TEMPERATURE SET POINT OF 55°F.
 - 2.2 IN HEATING, THE CONTROLLER SHALL MONITOR THE SUPPLY AIR TEMPERATURE AND MODULATE THE HOT WATER COIL TO MAINTAIN THE SUPPLY AIR TEMPERATURE SETPOINT OF 70°F.
 - 2.3 IN ECONOMIZER, THE UNIT SHALL RUN USING AN OUTSIDE AIR DRY-BULB BASED CONTROL STRATEGY.
3. FAN CONTROL:
 - 3.1 SUPPLY AIR FAN: THE CONTROLLER SHALL MONITOR DUCT STATIC PRESSURE AND MODULATE SUPPLY AIR FLOW THROUGH THE VFD TO MAINTAIN DOWNSTREAM STATIC PRESSURE SET POINT. SENSOR LOCATION AND SETPOINT SHALL BE DETERMINED BY THE TAB FIRM.
 - 3.2 EXHAUST AIR FAN: EXHAUST AIR SHALL MODULATE TO BE 200 CFM LESS THAN SUPPLY AIR TO MAINTAIN A POSITIVE STATIC PRESSURE FOR THE SPACE.
4. CONTROLLER:
 - 4.1 THE ONBOARD BACNET CONTROLLER SHALL COMMUNICATE WITH THE BAS TO ALLOW AT THE MINIMUM MONITORING AND CONTROL OF I/O POINTS LISTED HERE. THE SYSTEM SHALL BE ENABLED BY THE BAS. ALL CONTROL POINTS ON THE I/O POINTS LIST SHALL BE MONITORED AND CONTROLLED FROM THE GRAPHIC USER INTERFACE (GUI).
5. ALARMS:
 - 5.1 THE ONBOARD BACNET CONTROLLER SHALL TRANSMIT SYSTEM ALARMS TO THE GUI. THE ONBOARD CONTROLLER SHALL, AT A MINIMUM, TRACK ALARMS LISTED IN THE I/O POINTS LIST.

STATE OF TEXAS
RYAN EDWARDS
131050
LICENSED
PROFESSIONAL ENGINEER
08/09/2019



